

5.2 Simple linear regression as a warm-up

With a single regressor x and an intercept, the linear model is

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad i = 1, \dots, n, \quad (5.2.1)$$

where the x_i are known numbers and the errors satisfy the second-moment assumptions (L1)–(L3) of [Section 5.1](#): $E(\varepsilon_i) = 0$, $\text{Var}(\varepsilon_i) = \sigma^2$, and $\text{Cov}(\varepsilon_i, \varepsilon_j) = 0$ for $i \neq j$. The parameter β_1 is the **slope**, the difference in mean response between two values of x one unit apart. The **intercept** β_0 is the mean response at $x = 0$, which may or may not be a value of x that makes sense. Everything in this section is a special case of [Section 5.4](#). We work it out directly because the formulas are short, because they show the structure of the general answer, and because they are the ones used every day.

5.2.1 The least squares line

For a candidate line $b_0 + b_1 x$, the **residual** of case i is $y_i - b_0 - b_1 x_i$, the vertical distance from the data point to the line. The method of least squares chooses the line that minimizes the sum of squared residuals,

$$S(b_0, b_1) = \sum_{i=1}^n (y_i - b_0 - b_1 x_i)^2.$$

Other criteria are possible, for example the sum of absolute residuals (Chapter 45). Squared error is chosen because it leads to linear equations, because it matches the conditional mean of Proposition 5.1.1(c), and because its statistical properties under (L1)–(L3) can be worked out exactly, which is the business of this chapter.

We use the standard abbreviations

$$\begin{aligned} \bar{x} &= \frac{1}{n} \sum_i x_i, & \bar{y} &= \frac{1}{n} \sum_i y_i, & S_{xx} &= \sum_i (x_i - \bar{x})^2, \\ S_{xy} &= \sum_i (x_i - \bar{x})(y_i - \bar{y}), & S_{yy} &= \sum_i (y_i - \bar{y})^2. \end{aligned} \quad (5.2.2)$$

Two identities are used repeatedly: $\sum_i (x_i - \bar{x}) = 0$, and consequently $S_{xy} = \sum_i (x_i - \bar{x})y_i$ and $S_{xx} = \sum_i (x_i - \bar{x})x_i$. Note that $S_{xx} > 0$ unless all the x_i are equal.

Theorem 5.2.1 — Least squares for a straight line

Suppose the x_i are not all equal. Then $S(b_0, b_1)$ has a unique minimizer $(\hat{\beta}_0, \hat{\beta}_1)$, given by

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}. \quad (5.2.3)$$

It is the unique solution of the two **normal equations**

$$\sum_{i=1}^n (y_i - b_0 - b_1 x_i) = 0, \quad \sum_{i=1}^n x_i (y_i - b_0 - b_1 x_i) = 0. \quad (5.2.4)$$

Moreover, for all b_0, b_1 ,

$$\begin{aligned} S(b_0, b_1) &= S(\hat{\beta}_0, \hat{\beta}_1) + n[(b_0 - \hat{\beta}_0) + (b_1 - \hat{\beta}_1)\bar{x}]^2 \\ &\quad + S_{xx}(b_1 - \hat{\beta}_1)^2. \end{aligned} \quad (5.2.5)$$

Proof. S is a differentiable function of (b_0, b_1) , with

$$\frac{\partial S}{\partial b_0} = -2 \sum_i (y_i - b_0 - b_1 x_i), \quad \frac{\partial S}{\partial b_1} = -2 \sum_i x_i (y_i - b_0 - b_1 x_i).$$

A minimizer must make both derivatives zero, which gives [Eq. 5.2.4](#). The first normal equation divided by n says $\bar{y} - b_0 - b_1 \bar{x} = 0$, so $b_0 = \bar{y} - b_1 \bar{x}$. Substituting into the second, and using $y_i - \bar{y} - b_1(x_i - \bar{x})$ for the residual,

$$\sum_i x_i [(y_i - \bar{y}) - b_1(x_i - \bar{x})] = S_{xy} - b_1 S_{xx} = 0,$$

so $b_1 = S_{xy}/S_{xx}$. This is the only stationary point.

A stationary point need not be a minimum, so we check directly. Write $\hat{\varepsilon}_i = y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i$, $d_0 = b_0 - \hat{\beta}_0$ and $d_1 = b_1 - \hat{\beta}_1$. Then $y_i - b_0 - b_1 x_i = \hat{\varepsilon}_i - (d_0 + d_1 x_i)$, and

$$S(b_0, b_1) = \sum_i \hat{\varepsilon}_i^2 - 2d_0 \sum_i \hat{\varepsilon}_i - 2d_1 \sum_i x_i \hat{\varepsilon}_i + \sum_i (d_0 + d_1 x_i)^2.$$

The two middle sums vanish by the normal equations. For the last, $d_0 + d_1 x_i = (d_0 + d_1 \bar{x}) + d_1(x_i - \bar{x})$, and the cross term again sums to zero, so $\sum_i (d_0 + d_1 x_i)^2 = n(d_0 + d_1 \bar{x})^2 + d_1^2 S_{xx}$. This proves [Eq. 5.2.5](#). Both added terms are nonnegative, and since $S_{xx} > 0$ they are both zero only when $d_1 = 0$ and then $d_0 = 0$. So $(\hat{\beta}_0, \hat{\beta}_1)$ is the unique global minimizer. ■

The decomposition [Eq. 5.2.5](#) says more than minimality. It shows the level sets of S are ellipses centred at $(\hat{\beta}_0, \hat{\beta}_1)$, whose shape is set by n , $\bar{x}$ and S_{xx} . When $\bar{x}$ is far from zero the ellipses are long and thin, tilted along the direction where $b_0 + b_1 \bar{x}$ stays constant. Many lines then fit almost equally well, provided they pass near the centre of the data. This is a first look at the correlation between the estimated intercept and slope ([Corollary 5.5.4](#)).

The **fitted values** are $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$ and the **residuals** are $\hat{\varepsilon}_i = y_i - \hat{y}_i$. The minimum SSE = $\sum_i \hat{\varepsilon}_i^2$ is the **residual sum of squares** (also *error sum of squares*).

Proposition 5.2.2 — Properties of the fitted line

Under the conditions of [Theorem 5.2.1](#):

- (a) $\sum_i \hat{\varepsilon}_i = 0$, $\sum_i x_i \hat{\varepsilon}_i = 0$ and $\sum_i \hat{y}_i \hat{\varepsilon}_i = 0$;

- (b) the line passes through $(\bar{x}, \bar{y})$, and the mean of the fitted values is $\bar{y}$;
 (c) with $SST = S_{yy}$ and $SSR = \sum_i (\hat{y}_i - \bar{y})^2$,

$$SST = SSR + SSE, \quad SSR = \hat{\beta}_1^2 S_{xx} = \frac{S_{xy}^2}{S_{xx}},$$

and so $SSE = S_{yy} - S_{xy}^2/S_{xx}$;

- (d) if also $S_{yy} > 0$, the **coefficient of determination** $R^2 = SSR/SST$ equals r^2 , the square of the sample correlation $r = S_{xy}/\sqrt{S_{xx}S_{yy}}$.

Proof. (a) The first two are the normal equations Eq. 5.2.4. The third follows because $\hat{y}_i$ is a combination of 1 and x_i . (b) The first normal equation is $\bar{y} = \hat{\beta}_0 + \hat{\beta}_1 \bar{x}$, and averaging $\hat{y}_i = y_i - \hat{\varepsilon}_i$ gives $\bar{y}$. (c) By (b), $\hat{y}_i - \bar{y} = \hat{\beta}_1(x_i - \bar{x})$, so $SSR = \hat{\beta}_1^2 S_{xx} = S_{xy}^2/S_{xx}$. Also $y_i - \bar{y} = (\hat{y}_i - \bar{y}) + \hat{\varepsilon}_i$, and the cross term $\sum_i (\hat{y}_i - \bar{y})\hat{\varepsilon}_i$ vanishes by (a). (d) $R^2 = S_{xy}^2/(S_{xx}S_{yy}) = r^2$. ■

Part (c) is the simplest instance of the analysis of variance: the total variation of the response about its mean splits into a part along the fitted line and a part around it. Chapter 9 develops the general decomposition, and Section 6.7 interprets R^2 as a squared cosine.

5.2.2 Regression toward the mean

Divide S_{xy}/S_{xx} above and below by $n-1$. With the sample standard deviations $s_x = \sqrt{S_{xx}/(n-1)}$ and $s_y = \sqrt{S_{yy}/(n-1)}$, the slope is

$$\hat{\beta}_1 = r \frac{s_y}{s_x}, \quad \text{equivalently} \quad \frac{\hat{y} - \bar{y}}{s_y} = r \frac{x - \bar{x}}{s_x}. \quad (5.2.6)$$

In standard units, the fitted value is the regressor's standardized value multiplied by r . Since $|r| \leq 1$ (the Cauchy–Schwarz inequality, Proposition 1.1.1), a case that is two standard deviations above average in x is predicted to be *less* than two standard deviations above average in y , unless the points lie exactly on a line. This is **regression toward the mean**, and it gave the subject its name: Galton observed it in 1886 for the heights of parents and their adult children.

The effect is purely statistical and involves no mechanism. Regressing x on y instead gives the slope $r s_x/s_y$, which predicts x from y with the same shrinkage toward the mean in the other direction. The two fitted lines differ unless $r^2 = 1$. They answer different questions: the mean of y at given x , and the mean of x at given y . It is a common error to read regression toward the mean as evidence that extreme units become more ordinary over time, or that an intervention given to the most extreme units has worked.

5.2.3 A worked example

Example. (Forecasting December sea temperature)

The Niño 1+2 index is the mean sea surface temperature in a region of the Pacific off

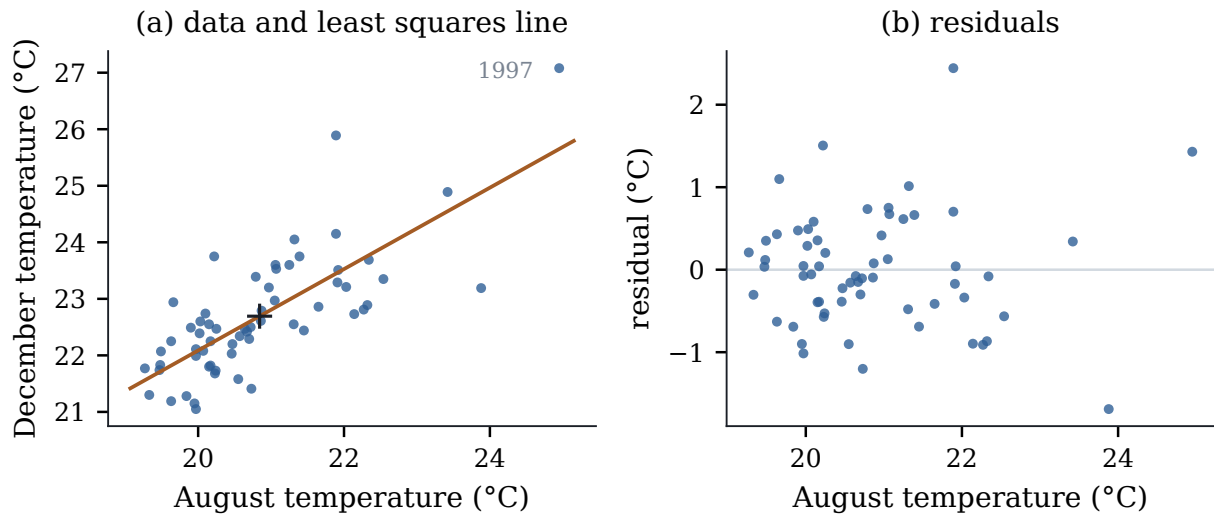


Figure 5.2.1: December against August sea surface temperature in the Niño 1+2 region, 1950–2010. (a) The data and the least squares line, which passes through the point of means (cross). (b) Residuals against the regressor. The El Niño year 1997 is the point at the far right of both panels.

Peru and Ecuador. It is high during El Niño events. We use the monthly values for the 61 years 1950–2010 and ask how well the August value predicts the December value four months later. Here x is the August temperature and y the December temperature, both in degrees Celsius. The summary statistics are

$$\begin{aligned}\bar{x} &= 20.843, & \bar{y} &= 22.693, \\ S_{xx} &= 77.801, & S_{xy} &= 56.008, & S_{yy} &= 70.380,\end{aligned}$$

so by [Theorem 5.2.1](#)

$$\hat{\beta}_1 = \frac{56.008}{77.801} = 0.7199, \quad \hat{\beta}_0 = 22.693 - 0.7199 \times 20.843 = 7.6887.$$

Each extra degree in August goes with 0.7199 of a degree more in the following December, on average. For an August temperature of 22°C the fitted December mean is 23.53°C. The intercept, the fitted December temperature after an August at 0°C, is meaningless: no August in the data is below 19°C.

The residual sum of squares is $SSE = 30.061$, and $R^2 = 0.5729$, the square of the correlation $r = 0.7569$. The slope is less than 1 partly because $r < 1$ and partly because December temperatures vary a little less than August ones: $s_y = 1.083$ against $s_x = 1.139$.

```
import numpy as np
import statsmodels.api as sm
sst = sm.datasets.elnino.load_pandas().data
x = sst["AUG"].to_numpy()           # August temperature
y = sst["DEC"].to_numpy()           # December temperature, same year
n = len(y)

xbar, ybar = x.mean(), y.mean()
Sxx = np.sum((x - xbar) ** 2)
Sxy = np.sum((x - xbar) * (y - ybar))
Syy = np.sum((y - ybar) ** 2)
```

```

b1 = Sxy / Sxx                # slope
b0 = ybar - b1 * xbar         # intercept
fitted = b0 + b1 * x
resid = y - fitted
print(f"n = {n}, slope = {b1:.4f}, intercept = {b0:.4f}")

```

[Run this code online](#)

```

print("sum of residuals      ", resid.sum())
print("sum of x * residuals  ", (x * resid).sum())
SSE = resid @ resid
R2 = 1 - SSE / Syy
r = Sxy / np.sqrt(Sxx * Syy)
print(f"R^2 = {R2:.4f}, r^2 = {r**2:.4f}")
print("numpy.polyfit agrees: ", np.polyfit(x, y, 1))

```

[Run this code online](#)

The residual plot in [Figure 5.2.1\(b\)](#) is the first diagnostic to draw after any fit. The residuals should show no pattern against x . Here there is no obvious curvature, but the spread is not uniform. The two warmest Augusts, in 1997 and 1983, both fell during strong El Niño events. The 1997 event was still building in December, while the 1982–83 event was over by then, and the two points sit well away from the line on opposite sides. The warmest of all, 1997 (August 24.95°C, December 27.08°C), has a residual of 1.430°C. Because it lies far from $\bar{x}$, it pulls on the slope: without it the slope would be 0.6214. Points with regressor values far from the mean have high *leverage*, a notion made precise in [Section 6.8](#) and Chapter 20.

5.2.4 Regression through the origin

Sometimes theory says the mean response is zero when $x = 0$, and the model $Y_i = \beta_1 x_i + \varepsilon_i$ is fitted without an intercept. Minimizing $\sum_i (y_i - b_1 x_i)^2$ gives one normal equation, $\sum_i x_i (y_i - b_1 x_i) = 0$, with solution

$$\tilde{\beta}_1 = \frac{\sum_i x_i y_i}{\sum_i x_i^2},$$

valid whenever some $x_i \neq 0$. The residuals now satisfy $\sum_i x_i \hat{\varepsilon}_i = 0$ but generally *not* $\sum_i \hat{\varepsilon}_i = 0$, and the decomposition of [Proposition 5.2.2\(c\)](#) about $\bar{y}$ fails. It holds instead about zero, as $\sum_i y_i^2 = \sum_i \hat{y}_i^2 + \sum_i \hat{\varepsilon}_i^2$, which is why software reports an “uncentred” R^2 for such fits. That number is not comparable with the R^2 of a model with an intercept ([Exercise \(A2\)](#)).

Dropping the intercept is a strong assumption. In [Example \(Forecasting December sea temperature\)](#), forcing the line through the origin gives slope 1.0877, a fit much worse than the line with an intercept: the data are nowhere near $x = 0$, and the constraint has nothing to do with how ocean temperatures behave.

5.2.5 Exercises

5.2.5.1 A. Check your understanding

Exercise A1.

For the five points $(x, y) = (0, 1), (1, 1), (2, 3), (3, 4), (4, 6)$, compute $\bar{x}, \bar{y}, S_{xx}, S_{xy}$, the least squares line, the residuals, and R^2 . Check the three identities of [Proposition 5.2.2\(a\)](#).

Solution. $\bar{x} = 2, \bar{y} = 3, S_{xx} = 10, S_{xy} = (-2)(-2) + (-1)(-2) + 0 + 1 \cdot 1 + 2 \cdot 3 = 13$. So $\hat{\beta}_1 = 1.3$ and $\hat{\beta}_0 = 3 - 2.6 = 0.4$. The fitted values are 0.4, 1.7, 3.0, 4.3, 5.6 and the residuals 0.6, -0.7, 0, -0.3, 0.4, which sum to zero. Also $\sum_i x_i \hat{\epsilon}_i = 0 - 0.7 + 0 - 0.9 + 1.6 = 0$. $S_{yy} = 4 + 4 + 0 + 1 + 9 = 18$, $\text{SSE} = 0.36 + 0.49 + 0 + 0.09 + 0.16 = 1.1$, and $R^2 = 1 - 1.1/18 = 0.939$, which equals $13^2/(10 \cdot 18)$.

Exercise A2.

For regression through the origin, show that $\sum_i y_i^2 = \sum_i \hat{y}_i^2 + \sum_i \hat{\epsilon}_i^2$, and give a small data set for which $1 - \sum_i \hat{\epsilon}_i^2 / \sum_i y_i^2$ is close to 1 while $1 - \sum_i \hat{\epsilon}_i^2 / S_{yy}$ is negative.

5.2.5.2 B. Practice**Exercise B1.**

Show that $\hat{\beta}_1 = \sum_i c_i y_i$ with $c_i = (x_i - \bar{x})/S_{xx}$, and that $\sum_i c_i = 0$, $\sum_i c_i x_i = 1$ and $\sum_i c_i^2 = 1/S_{xx}$. Write $\hat{\beta}_0$ as $\sum_i d_i y_i$ and find $\sum_i d_i$, $\sum_i d_i x_i$ and $\sum_i d_i^2$.

Solution. $S_{xy} = \sum_i (x_i - \bar{x})y_i$, so $\hat{\beta}_1 = \sum_i c_i y_i$. Then $\sum_i c_i = 0$, $\sum_i c_i x_i = \sum_i (x_i - \bar{x})x_i/S_{xx} = 1$, and $\sum_i c_i^2 = S_{xx}/S_{xx}^2 = 1/S_{xx}$. For the intercept, $\hat{\beta}_0 = \bar{y} - \bar{x}\hat{\beta}_1 = \sum_i d_i y_i$ with $d_i = 1/n - \bar{x}c_i$. Hence $\sum_i d_i = 1$, $\sum_i d_i x_i = \bar{x} - \bar{x} = 0$, and $\sum_i d_i^2 = \sum_i (1/n^2 - 2\bar{x}c_i/n + \bar{x}^2 c_i^2) = 1/n + \bar{x}^2/S_{xx}$. These are the coefficients that give the moments in Corollary 5.5.4.

Exercise B2.

Suppose all x_i are distinct. Show that $\hat{\beta}_1$ is a weighted average of the slopes $(y_j - y_i)/(x_j - x_i)$ of the lines through all pairs of points, with weights proportional to $(x_j - x_i)^2$. *Hint:* show that $\sum_{i < j} (x_j - x_i)(y_j - y_i) = nS_{xy}$ and $\sum_{i < j} (x_j - x_i)^2 = nS_{xx}$.

5.2.5.3 C. Going deeper**Exercise C1.**

The **least absolute deviations** line minimizes $\sum_i |y_i - b_0 - b_1 x_i|$. Show that some minimizing line passes through at least two of the data points (assume the x_i are not all equal). *Hint:* for fixed b_1 the best b_0 is a median of the $y_i - b_1 x_i$, and the objective is piecewise linear and convex in b_1 . Compare the effect of the 1997 point in [Example \(Forecasting December sea temperature\)](#) on this line and on the least squares line.